



- 1) If X is a random variable having Poisson distribution with mean λ , then show that $E(X^3) = \lambda^3 + 3\lambda^2 + \lambda$. [4 marks]
 b) A taxi cab company has 12 Ambassadors and 8 Fiats. If 5 of these taxi cabs are in the shop for repairs what is the probability that at least 3 of them are Ambassadors? [3 marks]
 Using normal distribution approximately find the find the probability of at least 60 heads if a fair coin is tossed 90 times. [3 marks]
- 2) Find the correlation coefficient for the two random variables X and Y whose joint density function is given as $f(x, y) = \begin{cases} x + y, & \text{if } 0 \leq x, y \leq 1 \\ 0, & \text{otherwise} \end{cases}$ [5 marks]
 b) If two random variables X and Y whose joint density function is given as $f(x, y) = \begin{cases} 4xy, & \text{if } 0 \leq x, y \leq 1 \\ 0, & \text{otherwise} \end{cases}$ [5 marks]
 and $Z = Y^2$ and $W = XY$, then find the joint density function of $g(z, w)$ of Z and W . [3 marks]
- 3) Measurements of the diameters of a random sample of 200 ball bearings made by a certain machine during one week showed a mean of 0.824 inch and a standard deviation of 0.042 inch. Find (i) 95%, (ii) 99% confidence intervals for the mean diameter of all the ball bearings. [4 marks]
 b) A sample of 20 brand A light bulbs showed a mean lifetime of 1400 hours and a standard deviation of 120 hours. A sample of 15 brand B light bulbs showed a mean lifetime of 1200 hours and a standard deviation of 80 hours. Find (i) 95%, (ii) 99% confidence limits for the difference of the mean lifetimes of the populations of brands A and B. [3+3 marks]
- 4) A new curing process developed for a certain type of cement results in a mean compressive strength of 5000 kilograms per square centimeter with a standard deviation of 120 kilograms. To test the hypothesis that $\mu = 5000$ against the alternative that $\mu < 5000$, a random sample of 50 pieces of cement is tested. The critical region is defined to be $x < 4970$. Find the probability of committing a type I error. Also, find the probability of committing a type II error for the alternative $\mu = 4960$. [3+3 marks]
 b) For testing:
 $H_0: \mu = 14$,
 $H_A: \mu \neq 14$,
 an $\alpha = 0.05$ level t-test is being considered. What sample size is necessary in order for the probability to be 0.1 of falsely failing to reject H_0 when the true population mean differs from 14 by 0.5? From a preliminary sample we estimate σ to be 1.25. [4 marks]
- 5)

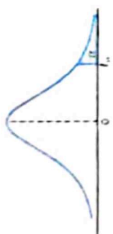
Height of Father (inches)	65	63	67	64	68	62	70	66	68	67	69	71
Height of Son (inches)	68	66	68	65	69	66	68	65	71	67	68	70

 [3+3+4 marks]
 Find regression lines i) $y = ax + b$ ii) $x = cy + d$ iii) $y = ax^2 + bx + c$

Normal Probability Table

z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995
3.3	0.9995	0.9995	0.9995	0.9996	0.9996	0.9996	0.9996	0.9996	0.9996	0.9997
3.4	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9997	0.9998

t-Distribution Probability Table



ν	α									
	0.40	0.30	0.20	0.15	0.10	0.05	0.025			
1	0.325	0.727	1.376	1.963	3.078	6.314	12.706			
2	0.289	0.617	1.061	1.386	1.886	2.920	4.303			
3	0.277	0.584	0.978	1.250	1.638	2.353	3.182			
4	0.271	0.569	0.941	1.190	1.533	2.132	2.776			
5	0.267	0.559	0.920	1.156	1.476	2.015	2.571			
6	0.265	0.553	0.906	1.134	1.440	1.943	2.447			
7	0.263	0.549	0.896	1.119	1.415	1.895	2.365			
8	0.262	0.546	0.889	1.108	1.397	1.860	2.306			
9	0.261	0.543	0.883	1.100	1.383	1.833	2.262			
10	0.260	0.542	0.879	1.093	1.372	1.812	2.228			
11	0.260	0.540	0.876	1.088	1.363	1.796	2.201			
12	0.259	0.539	0.873	1.083	1.356	1.782	2.179			
13	0.259	0.538	0.870	1.079	1.350	1.771	2.160			
14	0.258	0.537	0.868	1.076	1.345	1.761	2.145			
15	0.258	0.536	0.866	1.074	1.341	1.753	2.131			
16	0.258	0.535	0.865	1.071	1.337	1.746	2.120			
17	0.257	0.534	0.863	1.069	1.333	1.740	2.110			
18	0.257	0.534	0.862	1.067	1.330	1.734	2.101			
19	0.257	0.533	0.861	1.066	1.328	1.729	2.093			
20	0.257	0.533	0.860	1.064	1.325	1.725	2.086			
21	0.257	0.532	0.859	1.063	1.323	1.721	2.080			
22	0.256	0.532	0.858	1.061	1.321	1.717	2.074			
23	0.256	0.532	0.858	1.060	1.319	1.714	2.069			
24	0.256	0.531	0.857	1.059	1.318	1.711	2.064			
25	0.256	0.531	0.856	1.058	1.316	1.708	2.060			
26	0.256	0.531	0.856	1.058	1.315	1.706	2.056			
27	0.256	0.531	0.855	1.057	1.314	1.703	2.052			
28	0.256	0.530	0.855	1.056	1.313	1.701	2.048			
29	0.256	0.530	0.854	1.055	1.311	1.699	2.045			
30	0.256	0.530	0.854	1.055	1.310	1.697	2.042			
40	0.255	0.529	0.851	1.050	1.303	1.684	2.021			
60	0.254	0.527	0.848	1.045	1.296	1.671	2.000			
120	0.254	0.526	0.845	1.041	1.289	1.658	1.980			
∞	0.253	0.524	0.842	1.036	1.282	1.645	1.960			

ν	0.02	0.015	0.01	0.0075	0.005	0.0025	0.0005
1	15.894	21.205	31.821	12.133	63.656	127.321	636.378
2	4.849	5.643	6.965	8.073	9.925	14.089	31.600
3	3.182	3.896	4.541	5.017	5.841	7.453	12.924
4	2.999	3.298	3.747	4.088	4.604	5.598	8.610
5	2.757	3.003	3.365	3.634	4.032	4.773	6.869
6	2.612	2.829	3.113	3.372	3.707	4.317	5.959
7	2.517	2.715	2.998	3.203	3.499	4.029	5.408
8	2.449	2.634	2.896	3.085	3.355	3.833	5.041
9	2.398	2.571	2.821	2.998	3.250	3.690	4.781
10	2.359	2.527	2.764	2.932	3.169	3.581	4.587
11	2.328	2.491	2.718	2.879	3.106	3.497	4.437
12	2.303	2.461	2.681	2.836	3.055	3.428	4.318
13	2.282	2.436	2.650	2.801	3.012	3.372	4.221
14	2.264	2.415	2.621	2.771	2.977	3.326	4.140
15	2.249	2.397	2.602	2.746	2.947	3.286	4.073
16	2.235	2.382	2.583	2.724	2.921	3.252	4.015
17	2.224	2.368	2.567	2.706	2.898	3.222	3.965
18	2.214	2.356	2.552	2.689	2.878	3.197	3.922
19	2.205	2.346	2.539	2.674	2.861	3.174	3.883
20	2.197	2.336	2.528	2.661	2.845	3.153	3.850
21	2.189	2.328	2.518	2.649	2.831	3.135	3.819
22	2.183	2.320	2.508	2.639	2.819	3.119	3.792
23	2.177	2.313	2.500	2.629	2.807	3.104	3.768
24	2.172	2.307	2.492	2.620	2.797	3.091	3.745
25	2.167	2.301	2.485	2.612	2.787	3.078	3.725
26	2.162	2.296	2.479	2.605	2.779	3.067	3.707
27	2.158	2.291	2.473	2.598	2.771	3.057	3.689
28	2.154	2.286	2.467	2.592	2.763	3.047	3.674
29	2.150	2.282	2.462	2.586	2.756	3.038	3.660
30	2.147	2.278	2.457	2.581	2.750	3.030	3.646
40	2.123	2.250	2.423	2.542	2.704	2.971	3.551
60	2.099	2.223	2.390	2.504	2.660	2.915	3.460
120	2.076	2.196	2.358	2.468	2.617	2.860	3.373
∞	2.054	2.170	2.326	2.432	2.576	2.807	3.290